

Manideep Duggi

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Summary

Mechatronics and robotics systems engineer with experience in electromechanical systems, embedded control, sensor integration, validation, and root-cause debugging for autonomous robotic and industrial automation platforms. Skilled in C, Python, Linux, ROS2, telemetry/log analysis, hardware/software integration, PLC automation, PCB design, and field deployment support. Strong background collaborating with hardware, software, operations, and customer-facing teams to validate reliable systems in lab and real-world environments.

Technical Skills

Programming & Development: Python, C, C++ Fundamentals, Linux, Bash, Git, Docker

Systems Engineering & Validation: Requirements Decomposition, Root-Cause Analysis, Failure-Mode Analysis, Regression Testing, Edge-Case Testing, Hardware-in-the-Loop Testing, Technical Documentation

Electromechanical & Controls: Embedded Control, Sensors, Controllers, Drives, Actuators, Sensor Feedback, PID Control, State-Space Models, LQR/MPC Fundamentals, PLC Automation, Control Panels, PCB Design, Electro-Pneumatics

Robotics & Autonomy: ROS2, Nav2, Mobile Robot Navigation, Mobile Manipulation, Visual Servoing, Sensor Integration, Multi-Robot Systems

Planning & Algorithms: Path Planning, Task Allocation, Traffic Management, OMPL, RRT/RRT*, RVO

Sensors, Perception & Data: LiDAR, RGB-D Cameras, IMU, Encoders, OpenCV, YOLOv8, Mask R-CNN, MiDaS, Log/Telemetry Analysis

Deployment & Production Support: Hardware/Software Integration, Field Troubleshooting, Production Deployment Support, IoT Deployment, Performance Evaluation

Experience

Research Assistant

Jul 2025 – May 2026 | 11 months

RoboCare Lab, Worcester Polytechnic Institute — Multi-Robot Assistive Systems, ROS2, Navigation, Perception, Manipulation

Worcester, MA

- Built an end-to-end Pepper + Stretch 3 multi-robot assistive system integrating speech interaction, centralized task coordination, ROS2 navigation, RGB-D perception, object detection, grasp execution, handover, and user-facing task-state feedback into a working robot workflow.
- Developed coordination logic connecting user-facing task requests with robot-level execution, status tracking, navigation behavior, failure-state handling, confirmations, and real-time task updates, improving user task awareness from 15.8% to 84.6%.
- Designed and tested 6 preference-aware robot behavior modes derived from 9 user-preference traits, validating system behavior across interaction styles, task phases, and real-time execution states.
- Tuned and validated mobile manipulation behavior using RGB-D data, YOLOv8 segmentation, visual-servoing parameters, grasp approach thresholds, robot logs, and hardware-in-the-loop testing to improve object-fetch robustness across task phases.

Graduate Teaching Assistant

Mar 2026 – May 2026 | 3 months

ECE 2305, WPI — Communications & Networks, Lab Support

Worcester, MA

- Supported 70+ students through office hours, grading, feedback, and lab guidance involving Wireshark, client-server communication, networking concepts, and Python socket programming.

Robotics and Automation System Engineer

Sep 2023 – Jul 2024 | 11 months

Virya Autonomous Technologies — AMR Development, Field Deployment

Bengaluru, India

- Led planning, integration, testing, and root-cause debugging for 5+ AMR deployments using LiDAR, IMU, encoders, depth sensors, logs, and telemetry to diagnose localization drift, navigation failures, calibration issues, and edge cases.
- Supported production-style deployments by validating robot behavior in live industrial environments, documenting failures and fixes, coordinating with software, hardware, and operations teams, and improving field reliability and customer handover.
- Managed IoT-based remote firmware update and debugging workflows to maintain fleet performance across distributed customer sites.
- Optimized C/C++ firmware components for sensor integration and field deployment while collaborating with hardware teams to improve system reliability.

Engineer — Smart Solutions R&D

Maini Materials Movement Pvt. Ltd. — AMR, Industrial Automation

Aug 2022 – Aug 2023 | 1 year 1 month

Bengaluru, India

- Engineered AMR solutions for warehouse and industrial automation sites by translating customer workflows, traffic constraints, robot routes, maps, safety needs, and operational tradeoffs into deployment-ready plans.
- Conducted 20+ industrial site assessments to evaluate automation feasibility, material movement workflows, facility layouts, safety requirements, and deployment constraints.
- Supported AMR system integration, field testing, commissioning, and customer handover for smart material-handling automation solutions.
- Collaborated with mechanical, electrical, software, and operations teams to troubleshoot deployment issues and improve system performance in real-world environments.

Automation Engineer Trainee

Tranqvolt Robotics and Festo — PLC, PCB Design, Electro-Pneumatics

Dec 2021 – Jul 2022 | 8 months

Hyderabad, India

- Built PLC-based automation systems, control panels, PCB layouts, and electro-pneumatic solutions for production-line applications across 5+ industrial sites.
- Coordinated with electricians and technicians during wiring, installation, testing, debugging, and commissioning to deliver working industrial automation systems.

Selected Projects

Multi-Robot Motion Planning and Simulation

ROS2 / Python / OMPL

- Developed a multi-robot motion planning framework in ROS2 using C++ and Python with OMPL integration for collision-aware navigation.
- Implemented and evaluated RRT, RRT*, KPIECE, prioritized multi-robot planning, and RVO for collision-aware robot navigation.
- Compared planner performance using simulation-based validation, collision checking, runtime analysis, path-quality metrics, and edge-case testing.
- Integrated simulated LiDAR and IMU sensor models and conducted hardware-in-the-loop testing to validate planner robustness under sensor noise.

Reinforcement Learning-Based Manipulation: Franka Cube Stacking in Genesis

Python / PPO

- Trained cube-stacking manipulation policies using PPO with imitation learning and behavior cloning to improve task stability, sample efficiency, and policy convergence.
- Designed reward shaping and evaluation metrics including stack height, object alignment, task success rate, and motion consistency to measure manipulation performance.
- Performed hardware-in-the-loop simulation with force-torque and vision sensors to validate manipulation policies under realistic dynamics.

Autonomous Vehicle Safety and Lane-Change Control

MATLAB / Controls / Vehicle Dynamics

- Modeled autonomous lane-change scenarios using vehicle dynamics, state-space representation, LQR/MPC-style control concepts, and safety constraints for behavior analysis.

Einstein Vision — 2D to 3D Scene Understanding with Tesla-Style Dashboard

Computer Vision / Perception

- Built a YOLOv8, Mask R-CNN, MiDaS, and camera-intrinsics-based perception pipeline with a Tesla-style dashboard for visualizing detected objects, depth estimates, scene understanding outputs, and real-time system status.

Research

Socially Mediated State Externalization in Multi-Robot Assistive Systems

IROS 2026, Under Review

- Co-authored research on transparency, trust, and user awareness in assistive multi-robot systems through social robot-based task-state communication.

Education

Worcester Polytechnic Institute (WPI)

M.S. Robotics, Automation and Mechatronics Engineering | GPA: 3.60/4.00

2024 - 2026

Worcester, MA, USA

SRM Institute of Science and Technology

B.Tech. Mechatronics Engineering | GPA: 8.14/10

2018 - 2022

Chennai, TN, INDIA